



Executive Architecture Summary

BloodSync Android application me Google Firebase aur Play Services ki API keys client-side par **Authentication, Real-Time Cloud Firestore, Emergency SOS Radar, Push Notifications (FCM)** aur **Google 1-Tap Sign-In** ko operate karne ke liye use hoti hain. Yeh keys google-services.json ke dwara Gradle build system se automatically compile-time par inject hoti hain.

Mukhya Credentials & Keys

1. Google / Firebase Mobile API Key

AIzaSyBTuYi0hGnq6lculx8c4HsjPF7ocIM1dU

Source: app/google-services.json

Scope: Firebase Auth, Firestore Realtime Database, Cloud Messaging (FCM).

2. Google OAuth Web Client ID

1044746314963-01csrn69c2eut7dnhqcr67oml08244n...

Source: AuthScreen.kt (GoogleSignInOptions)

Scope: 1-Tap Google Sign-In & ID Token Authentication.

App Ke Wo Features Jahan API Keys Ka Use Ho Raha Hai

#	Feature / Screen	Kaam (Detailed Purpose)	App Code File
1	Donor Registration	Naye donor ka name, email, password aur credentials Firebase Auth cloud me create karna.	AuthScreen.kt BloodSyncRepository.kt
2	User Login (Sign-In)	Registered donor ka email & password verify karke authenticated session start karna.	AuthScreen.kt BloodSyncRepository.kt
3	"Continue with Google"	1-Tap Google sign-in prompt kholna aur bina password direct secure login karwana.	AuthScreen.kt (Line 143)
4	Self-Service Password Reset	User ke registered email par secure password reset link bhejna.	AuthScreen.kt BloodSyncRepository.kt
5	Donor Directory & Search	A+, B+, O+, AB+ blood groups ke verified donors ki list Firestore cloud se fetch karna.	FirebaseSyncService.kt DonorsDirectoryScreen.kt
6	Emergency SOS Broadcast	Urgent blood requirement create karna aur nearby volunteers ko real-time broadcast karna.	EmergencyRequestScreen.kt FirebaseSyncService.kt
7	Live Emergency Tracking	Emergency blood request ka real-time status, accepted donor aur location update dekhna.	EmergencyLiveTrackingScreen.kt
8	User Profile & DOB Sync	User ka Name, Blood Group, Age, Date of Birth (DOB), Phone aur City cloud par save karna.	FirebaseSyncService.kt ProfileScreen.kt
9	Push Notifications (FCM)	Urgent blood requests par donor ke phone par high-priority sound aur pop-up notification bhejna.	BloodSyncFirebaseMessagingService.kt
10	Hospital & Blood Bank Directory	24/7 verified blood banks aur unka live stock status cloud database se load karna.	BloodSyncRepository.kt BloodBanksScreen.kt
11	Donation Certificates & History	User ke lifetime blood donation records aur digital certificates cloud se retrieve karna.	FirebaseSyncService.kt ProfileScreen.kt
12	Web Admin Dashboard	Computer browser me pura BloodSync network, donors aur requests live monitor karna.	bloodsync-dashboard.html